

# VIGNESH R

vignesh01ak@gmail.com6383527141LinkedlnPortfolioLeetcodeGithub

## SUMMARY

Aspiring Full Stack Developer skilled in building scalable web applications using React.js, JavaScript, and Python, with experience in REST API development, authentication (JWT, Firebase), and database management (MySQL). Strong foundation in data processing, validation, and workflow optimization, with a focus on performance and user-centric solutions.

## WORK EXPERIENCE

Centillion Networks Private LimitedNov 2024 – Dec 2025

- Worked on telecom network design and landbase mapping for infrastructure planning. Ensured accurate representation of network layouts and connectivity
- Performed detailed quality checks and validation of network design data, identifying inconsistencies and improving overall data accuracy and reliability
- Analyzed and processed large volumes of structured data, improving data accuracy and workflow efficiency.
- Collaborated with cross-functional teams to maintain data consistency, streamline workflows, and support timely delivery of network projects
- Assisted in optimizing workflow processes by reducing manual errors and improving data handling efficiency through structured validation techniques

Web Development Intern - Prodigy InfotechJan 2025-Feb 2025

Projects: [StopWatch](#) | [TicTactoe](#) | [Weather app](#)

- Developed 3 responsive web applications (Stopwatch, Tic-Tac-Toe, Weather App) using HTML5, CSS3, and JavaScript, improving UI responsiveness and cross-device compatibility.
- Integrated OpenWeather RESTful API to fetch and render real-time weather data with dynamic UI updates.

## PROJECTS

Workflow Management Dashboard | React.js, Firebase Auth, RESTful API, JavaScript, Git

- Designed modular frontend architecture with reusable components and dynamic role-based navigation.
- Implemented Context API for global state management, eliminating prop drilling across nested components.
- Implemented secure routing using React Router to enforce role-based access control.
- Integrated REST APIs to enable real-time project and employee data processing.

Hand Gesture and Speech Recognition | Python, OpenCV, Speech Recognition

- Built a real-time multimodal system combining hand gesture recognition (OpenCV) and speech-to-text processing for voice command execution
- Implemented computer vision pipeline to detect and classify hand gestures via live camera; integrated voice commands for dynamic execution.
- Optimized low-latency real-time processing and designed modular architecture for scalable integration — enabling touchless, natural user interaction.

Ethical Keylogger — Security Research | Python

- Engineered a Python-based keylogger achieving 40% improvement in data capture efficiency with full AES encryption of captured data.
- Implemented automated keyword-based alerting, reducing incident response time by 30% and cutting false alarms by 15%

## TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), SQL  
Frontend: HTML5, CSS3, React.js  
Backend: Django REST Framework  
Databases: MySQL, Firebase  
Tools: Git, GitHub, Visual Studio Code, Netlify, Postman, Vercel.

## EDUCATION

HOLY PRINCE MATRICULATION HIGHER SECONDARY SCHOOL

- |        |     |           |
|--------|-----|-----------|
| • HSC  | 81% | 2019-2020 |
| • SSLC | 86% | 2017-2018 |

KCG COLLEGE OF TECHNOLOGY

- |                                             |      |           |
|---------------------------------------------|------|-----------|
| • Bachelor of Engineering(Computer Science) | 8.02 | 2020-2024 |
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## CERTIFICATE

- Introduction to Python - Coursera
- Cyber Security Essentials - Cisco Networking Academy